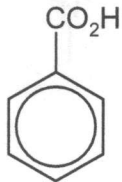
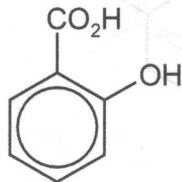


26 The pK_a values of benzoic acid and the hydroxybenzoic acids are shown.



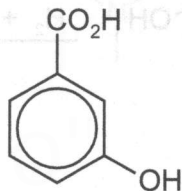
benzoic acid

$pK_a = 4.20$



2-hydroxybenzoic acid

$pK_a = 2.98$



3-hydroxybenzoic acid

$pK_a = 4.08$



4-hydroxybenzoic acid

$pK_a = 4.58$

Why is 2-hydroxybenzoic acid so much more acidic than 4-hydroxybenzoic acid?

- A A pair of its molecules are hydrogen-bonded to each other.
- B Its conjugate base is stabilised by internal hydrogen bonding.
- C Its hydroxy group has electron-withdrawing properties.
- D The negative charge of its conjugate base is spread over two oxygen atoms.